Can AI-Human interaction drive sound change? Insights from phonetic accommodation in Hong Kong Cantonese [ŋ-]-[Ø-] onset merger

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Artificial Intelligence(AI)-generated voices are everywhere

Wide application in different fields

- Second language acquisition (e.g., Song et al., 2022)
- Speech therapy tools (e.g., Deka et al., 2025)
- Mental health interactions (e.g., Liu et al., 2025)
- In-home voice assistants (e.g., McLean & Osei-Frimpong, 2019)
- Customer services (e.g., Wang et al., 2023)
- Advertisements (e.g., Wang et al., 2024)
- …

→ 🤖 **Human-AI interaction is the new norm**

Speech is modulated differently in Human-AI interactions

Different phonetic realizations compared to human-human interactions

- *Duration* (e.g., Cohn et al., 2024; Chen et al., 2026)
- *Spectral Qualities* (e.g., Cohn & Zellou, 2021; Chen et al., 2026)
- *F0* (e.g., Bradshaw et al., 2023, Cohn et al., 2022, Hong et al., 2026)
- *Intensity* (e.g., Raveh et al., 2019)
- …

→ **Primarily focus on immediate effects (i.e., within the interactions)**

→ **Could there be any carryover effects (i.e., after the interactions)?**

Could there be any carryover effects after interactions?

Sound Change

- Allows us to systematically examine potential **carryover** effects
- **Before** AI era…
  - Human listeners and speakers primarily contribute changes to their speech community
- **Now we are now in** the AI era…
  - AI-generated speech becomes an additional input to the existing community

→ **Regardless, listeners remain human**

→ **What makes human listeners crucial to sound change?**

We are not perfect listeners 👂 + tend to imitate 🗣️

= Phonetic Accommodation

- Step 1. Listeners: Exposed to a variant
- Step 2. Listener-turned-Speakers: Accommodate 🗣️ the perceived variant (intended or distorted) (Ohala, 2012)
  - Convergence (i.e., phonetic imitation) is the most reported type of accommodation to human voices

- Accommodation contributes to sound change in progress (Garrett & Johnson, 2013)
  - Accommodation ~ a mini-sound change

→ Do listeners also accommodate AI voices?

→ If so, could AI voices influence sound change differently?

Positioning AI voice in sound change

RQ: Does AI have the potential to contribute to sound change?

- Specifically, we examine phonetic accommodation (i.e., a mini-sound change):
- 1. Do accommodation patterns differ **between AI and human voices**?
- 2. Do accommodation patterns remain **after shadowing**?

Positioning AI voice in sound change (Cont.)

🎯 Target feature: Hong Kong Cantonese (HKC): [ŋ-]-[Ø] onset merger

- E.g., 餓 **[ŋɔː^{22}]–[ɔː^{22}]** ‘hungry’
- **Why this sound change?**
  - 1. 🕰️ A bidirectional merger **remains unsettled after 100 years**
    - → Can AI voices **alter its** 📈 **trajectory** 📉 ?
  - 2. The only HKC sound change feature with **a reported loss of the production-perception link** 🔁 (Cheng at al. 2022)
    - → Does 👂 **listener-based accommodation still operate**?

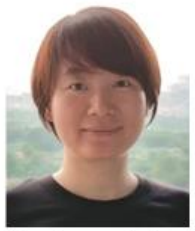
Method - ⭐We trained our own AI speech

Generated using DurIAN (Duration Informed Attention Network) TTS model

- 2 phonetically-trained, native HKC female RA
- 4000 Cantonese sentences each

(Thanks to our engineering team!)


Tan Lee


Yusheng Tian

Stimuli	Meaning	Shadowing Variant Played
我	'I'	[ŋɔː^{23}]
餓	'hungry'	[ŋɔː^{22}]
外	'outside'	[ɔːi^{22}]
哀	'sorrow'	[ɔːi^{55}]
愛	'love'	[ɔːi^{33}]
曖	'ambiguous'	[ɔːi^{25}]

Why customized AI voice?

- Voice manipulation was **NOT** considered previously
- → **Control the variants produced**, esp. in the study of sound change

Method - Exposure to both human and AI voices

Participants (n = 33)

- 21.52 ± 3.76; 17F & 16M
- **Between-subject** factors:
  - Face (Human / Avatar)
  - Model talker identity (RA1 / RA2)
- **Within-subject** factor:
  - Voice (Human / AI)
- **Order** of **voice** is **counterbalanced**

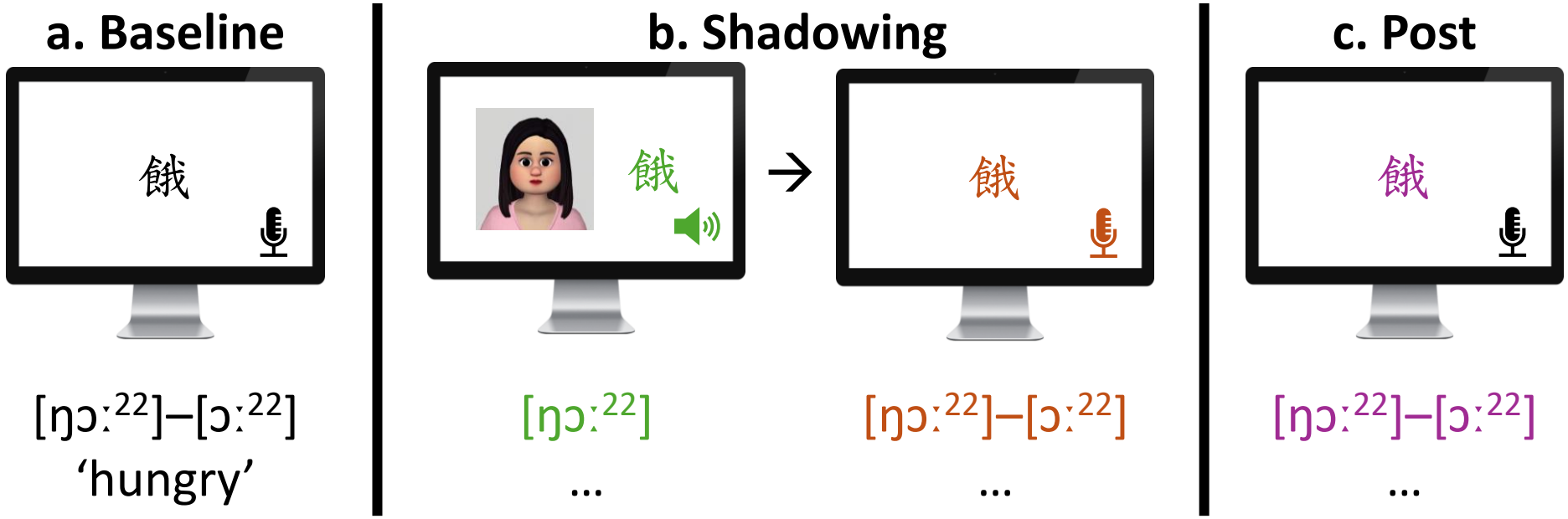
Voice \ Stimuli	餓 [ŋɔː^{22}] ‘hungry’	愛 [Øɔːi^{33}] ‘love’
👩 **Human**		
🤖 **AI-generated**		

→ Closely resemble human voices
→ Subtle perceivable diff. due to synthesis

Method - Assess Accommodation using shadowing task

Shadowing task with a model talker

- 👂 A listener-turned-speaker-based experimental technique



- → Similar mechanism as how a change is initiated and spread (Garrett & Johnson, 2013)

Measurement - Temporal Adjustment = Accommodation

Temporal adjustment as operationalization of **phonetic accommodation**

E.g., Convergence (i.e., become model talker-like)

1. Listen to **shorter** vowel proportions
2. Produce **shorter vowel proportions** when becoming a speaker

🗣️ **Baseline**	ŋ	ɔː
👂 **Listen to**	ŋ	ɔː
🗣️ **During shadowing**	ŋ	ɔː
🗣️ **After shadowing**	ŋ	ɔː

Analysis - Linear Combination

~~Difference-in-difference (DID) approach: classic yet biased~~ (Cohen Priva & Sanker, 2019; MacLeod, 2021)

2 LMMs:

Shadowing / Post-Shadowing ~

Baseline +

ModelTalker*Voice*Face +

$ModelTalkerID_{(RA1/2)}$ +

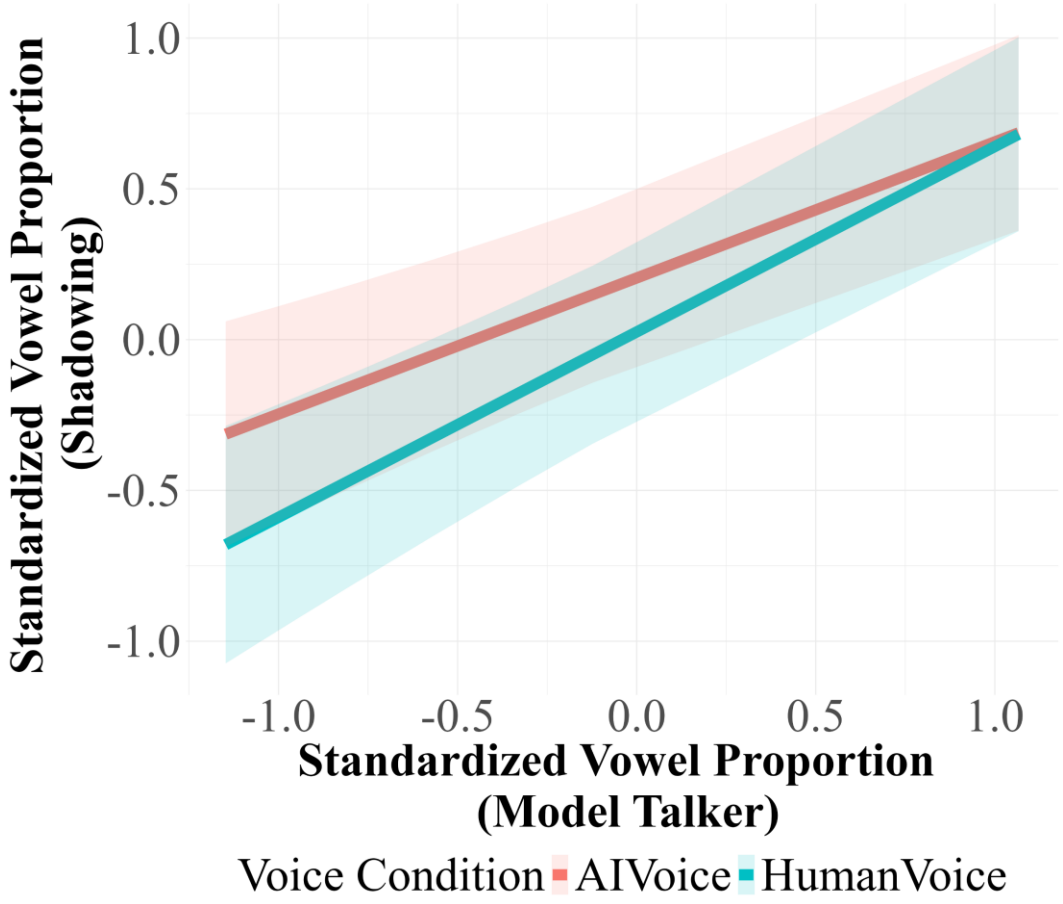
(1|ParticipantID) + (1|Character) + (0 + ModelTalker| ParticipantID)

Diagnostics

✅ Multi-collinearity, Singularity, Homoscedasticity, and Normality of residuals

Results - (1) Convergence found during shadowing

= Longer prop. in the model talker → Longer prop. in participants, VV



Convergence depends on the voice source

- Stronger convergence on the human voices

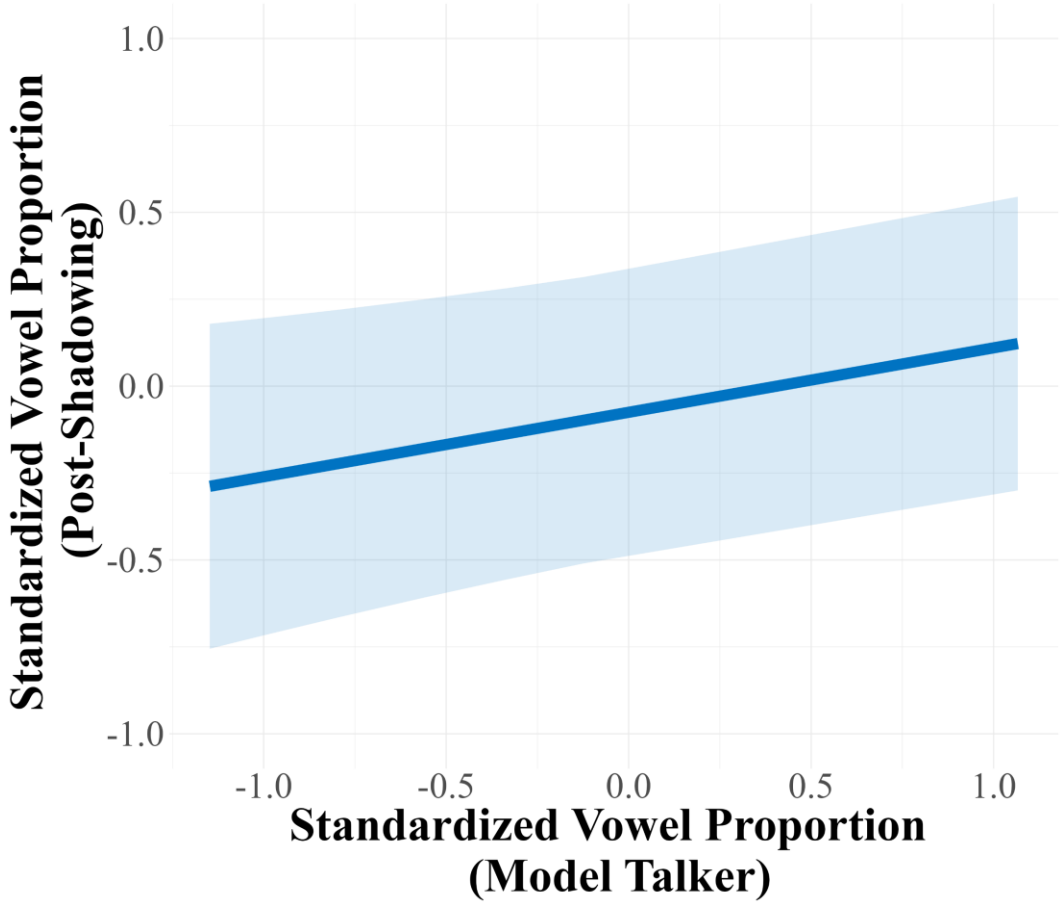
(β = - .25, SE = .07, 95% CI [-0.38, -.11], t(286) = -3.55, p < .001)

- → Participants were **more strongly influenced by human speech** during shadowing
- No other main effects or interactions were sig.

Immediate effects ✅ → Any carryover effects ❓

Results - (2) Convergence retained after shadowing

= Longer prop. in the model talker → Longer prop. in participants, VV



No sig. effects of voice source

(F(1, 290.54) = 0.03, p = .87)

- → No sig. diff. in convergence btw. voices
- → ✅ **Participants' tendency to imitate remained**
- No other main effects or interactions were sig.

Discussion - AI's influence on speech performance

During - Sig. diff. between AI and Human model talkers

- Align with previous finings on Human-AI interaction (i.e., modifications within interactions)

After - No diff. between talkers, yet overall convergence remained

- Comparable effect of AI and human voices over time in affecting speech behaviour

Increasing exposure to AI voices from various sources in recent years

- Multiple short-term carryover accommodations

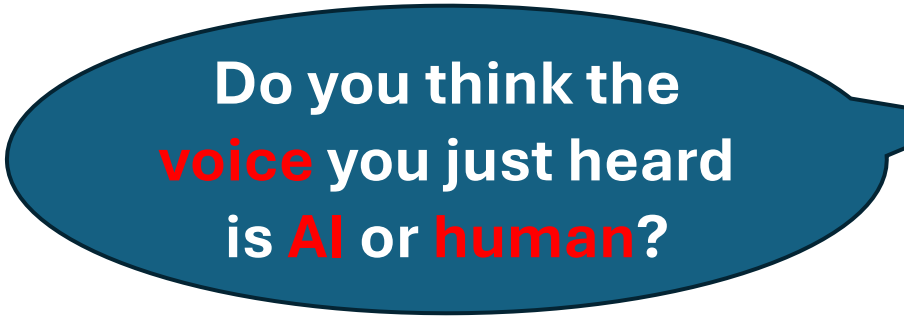
→ A potential impact on sound change

To speech engineer: Variants generated matter!

Discussion - Accommodation is more than just automatic

A socially-mediated behaviour

- Whether a sound is adopted after listening depends on the listener's **social/subjective evaluations of model talkers**
  - E.g., attractiveness, dialect, social status, and other



Face (between) / Voice (within)		Human Face (n = 16)	AI Face (n = 17)
Correctly Identify		8 (50.0%)	5 (29.4%)
Wrongly Identify as	Both AI	3 (18.8%)	4 (23.5%)
	Both Human	5 (31.3%)	8 (47.1%)

- More than 60% participants identified the voice sources **incorrectly**
  - Such awareness could further affect speech performance
  - Yet sample size in each subgroup now is not suitable for any generalization

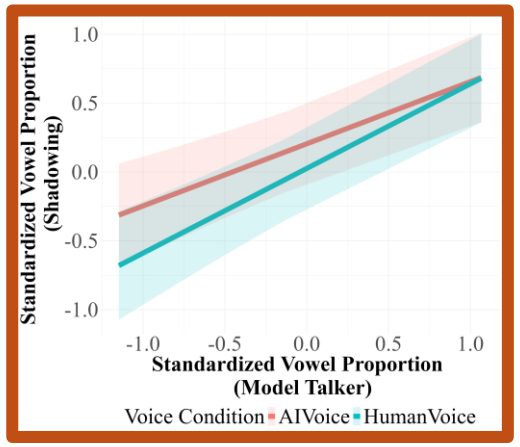
→ More data is needed to address this question

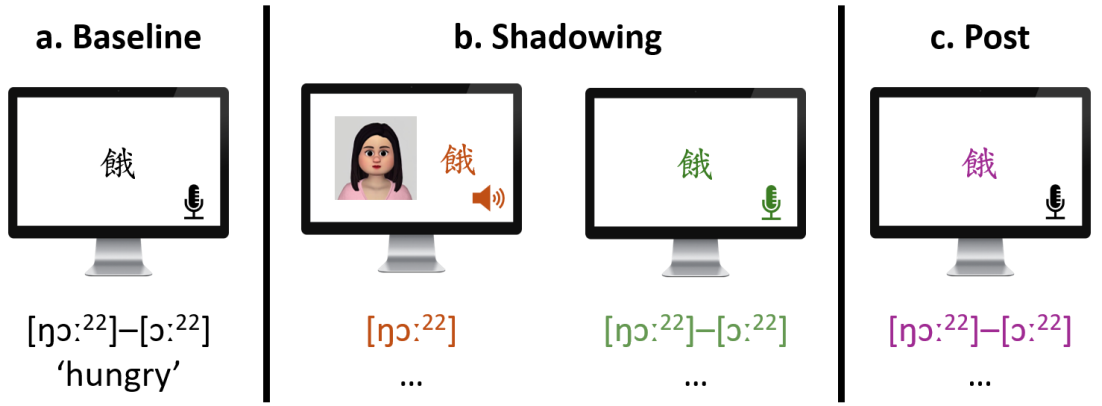
Conclusion - AI has the potential to contribute to sound change?

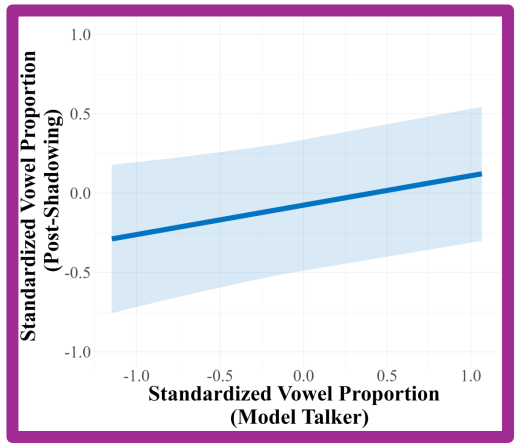
Yes, mini-sound change was found!

- **Convergence (phonetic imitation) during and after exposure**

Stronger influence from human → No sig. diff. btw. (yet imitation retains)





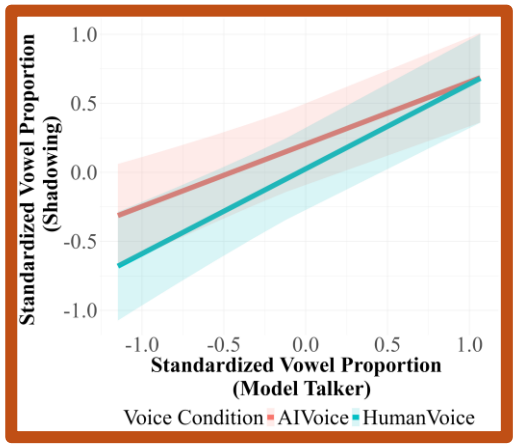


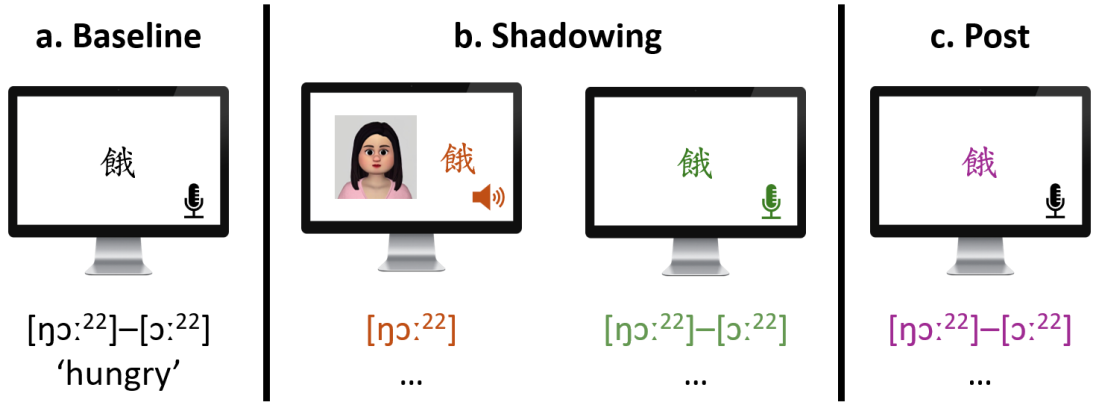
3 Takeaways

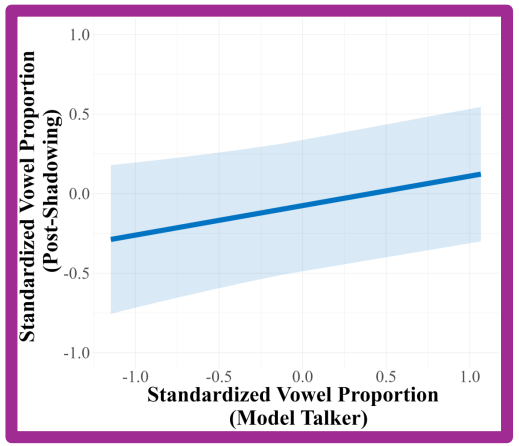
→ Both human and **AI voices** can **influence speakers' speech after exposure**

→ **Repeated carryover effects** may accumulate to **long-term speech changes**

→ **AI-generated variant** has the potential to **contribute to sound change**







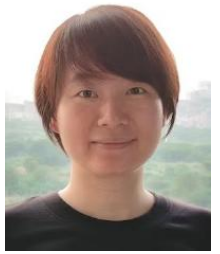
Acknowledgements

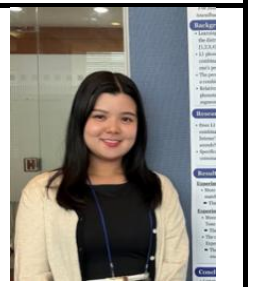
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Thank you and comments are welcome!

Slides are available at https://mansonwongcm.github.io/projects.html

